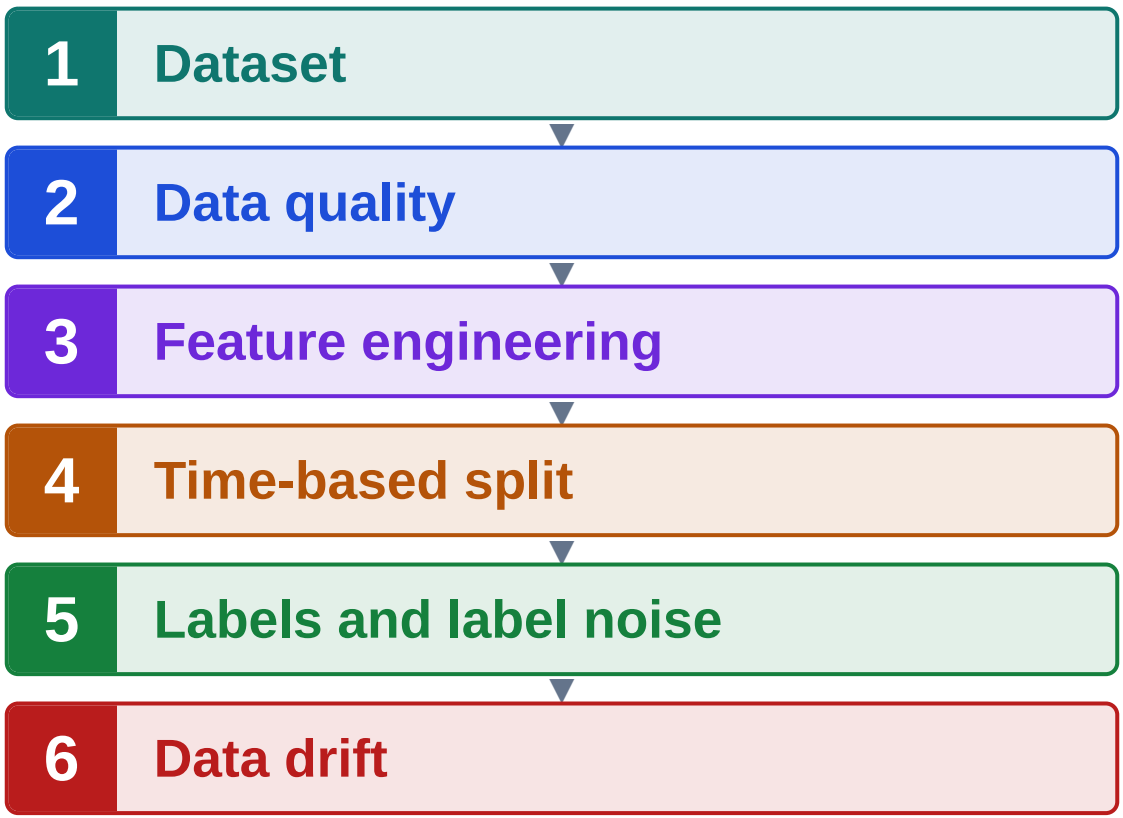
Lecture 4

# Preparing Data for Analytical Models

Data quality · features · time-based split · drift

# Lecture outline



1

# Dataset

what the model learns from

# Basic dataset concepts

| Concept | Meaning | Example |
|---|---|---|
| **Dataset** | A table from which the model takes examples for training and evaluation | All objects over 56 days of observation |
| **Object** | One row of the dataset: what the model makes a decision about | A source address over a 5-minute window |
| **Feature** | A numeric characteristic of an object computed from the log | Number of login attempts in the window |
| **Label** | The known answer for an object: attack or normal activity | “attack: password guessing” |
| **Subset** | A part of the dataset with a specific role | Training, validation, test |

> Use only what will also be known at the moment of real-world use

# From log records to an object

| Time | Address | Account | Outcome |
|---|---|---|---|
| 09:00:04 | 203.0.113.7 | anna | failure |
| 09:00:09 | 203.0.113.7 | anna | failure |
| 09:00:15 | 203.0.113.7 | anna | failure |
| 09:00:21 | 203.0.113.7 | anna | failure |
| 09:02:40 | 198.51.100.5 | boris | success |

→

**Object**

address 203.0.113.7, window 09:00–09:05

| | |
|---|---|
| Attempts | **4** |
| Failure share | **1.0** |
| Distinct accounts | **1** |
| Label | **attack** |

An object is a dataset row: numeric features and a label instead of log records

# Specifics of security data

**Class imbalance**

attacks are a fraction of a percent of objects

**Time dependence**

events of one attack come in a row

**Drift**

behavior of users and attackers changes

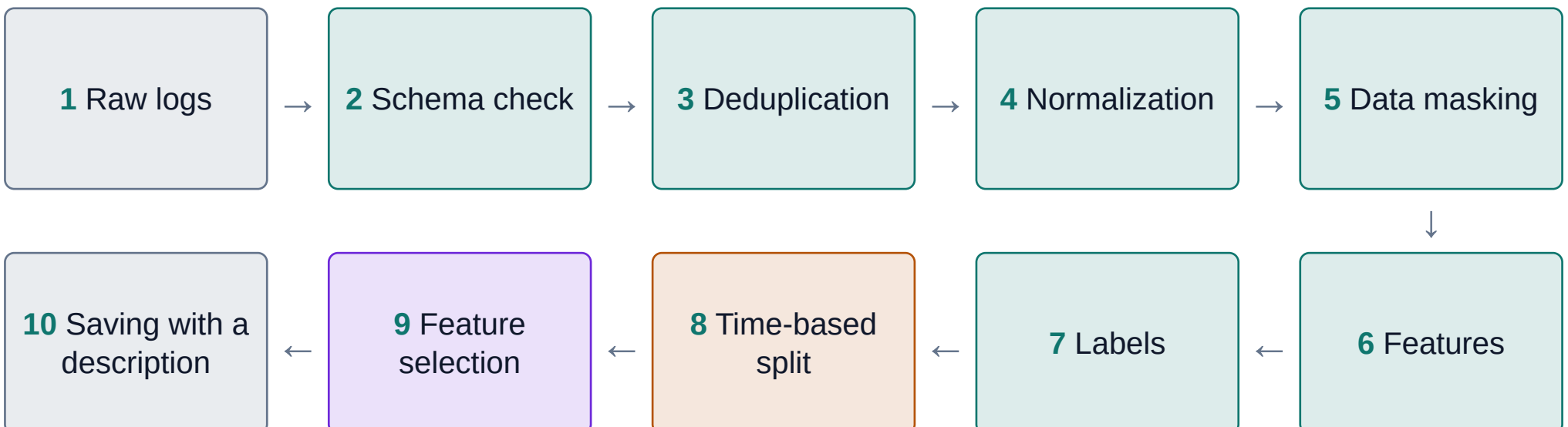
**Label noise**

some attacks went undetected and are labeled normal

**Adversarial behavior**

the attacker adapts to defenses

# Data preparation pipeline



# 2

# Data quality

check before training

# Typical problems

**Missing values**

biased estimates

**Duplicates**

inflated quality

**Gaps in the stream**

silence is taken as normal

**Different formats**

parsing and event ordering errors

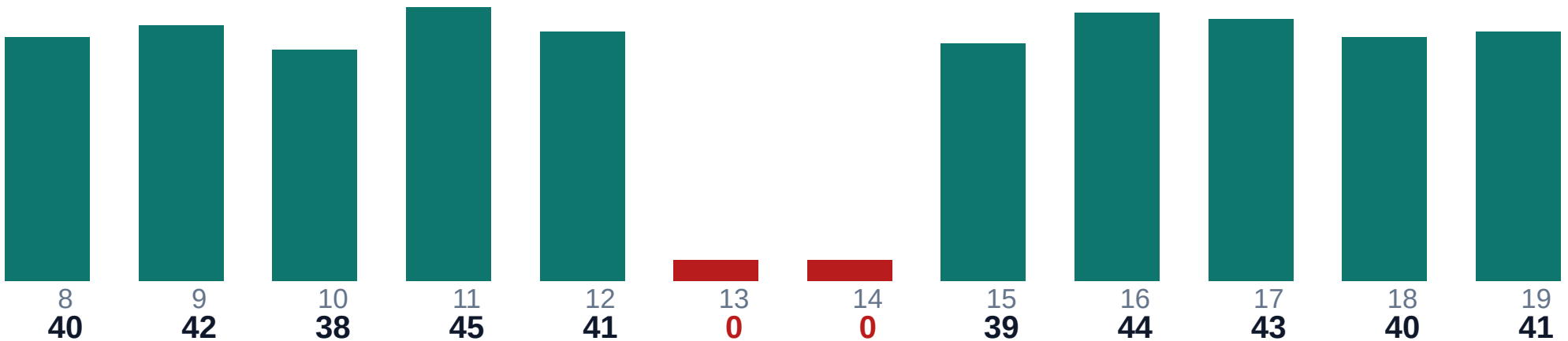
**Wrong labels**

the model learns from mistakes

**Personal data**

risk of dataset leakage

# A gap in the stream looks like silence




13–14 h: the source was silent while an attack was going on — the data says “normal”

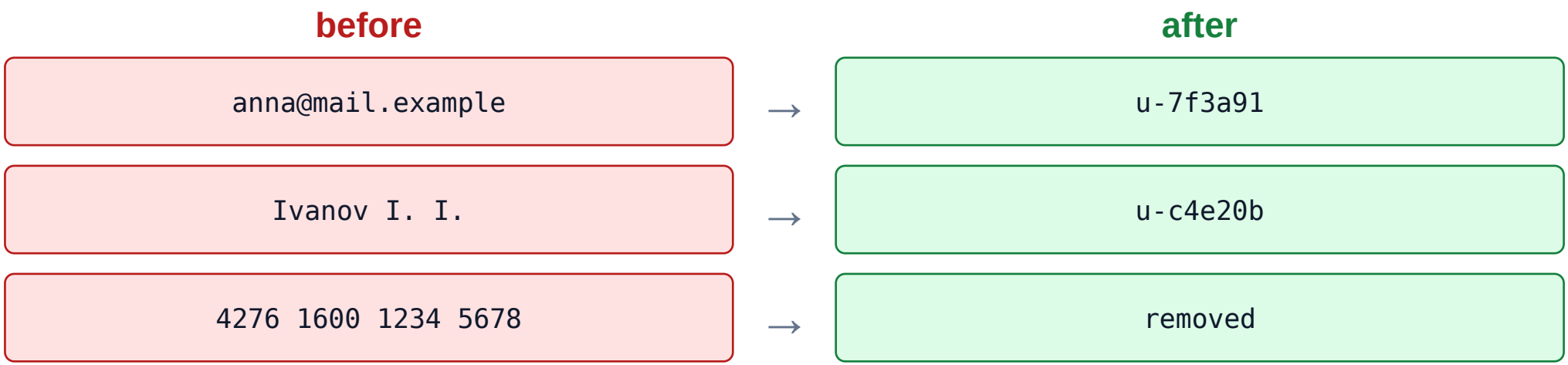
Zero and abnormally low hours are flagged and checked

# Quality checks

| | |
|---|---|
| **Record schema** | required fields, types, allowed values |
| **Duplicates** | by record ID or by a combination of time, source, type |
| **Continuity** | number of records per hour; zeros are flagged |
| **Consistency** | time in UTC, one address format, one letter case |
| **Time order** | timestamps not in the future and not outside the period |

> Failing any check stops the pipeline

# Masking before feature engineering




The pseudonym is stable: per-user aggregates can be computed, identity cannot be established without the key

# 3

# Feature engineering

what the model sees

# Three kinds of features

## Event-level

a characteristic of a single record

hour of day, failure reason, client application

## Aggregate

entity statistics over a window

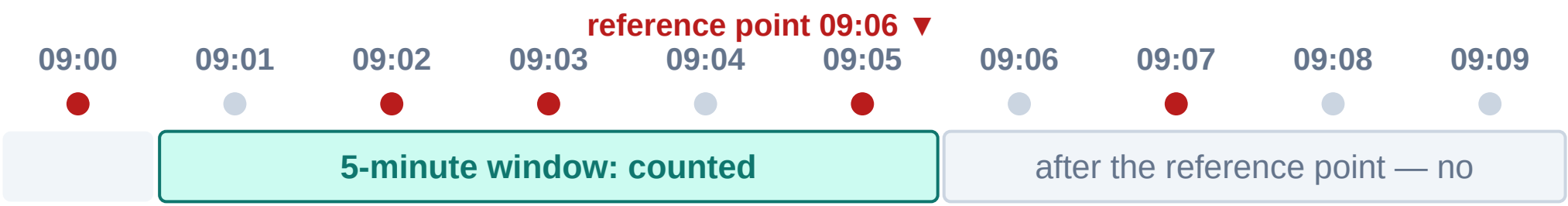
number of attempts, failure share, number of distinct accounts

## Statistical

deviation from the usual level

attempts relative to the 7-day mean

# Window and reference point




The window ends at the reference point: future events do not exist yet in real operation

# Four objects — four sets of numbers

| Object (5-minute window) | Attempts | Failure share | Distinct accounts | Label |
|---|---|---|---|---|
| A · user mistyped the password | 2 | 0.50 | 1 | **normal** |
| B · guessing one account | 8 | 1.00 | 1 | **attack** |
| C · password spraying | 9 | 1.00 | 9 | **attack** |
| D · office behind a shared address | 40 | 0.08 | 25 | **normal** |

No single feature separates all cases: failure share tells C from D, distinct accounts tell B from C

# Feature selection and documentation

## Selection methods

- Drop highly correlated features (|r| > 0.95)
- Drop near-constant features
- Mutual information with the label
- Permutation importance

## Feature description

**Name:** `failure_share_5m`
**What:** share of failures among attempts in 5 minutes
**Range:** a number from 0 to 1
**Source:** field `outcome`, type `auth.login`
**Window:** 5 minutes before the reference point
**Edge case:** with 0 attempts — value 0

The decision which features to keep is made on the training set only

# 4

# Time-based split

protection against data leakage

# Leakage of data from the future

**Random split**

one attack ends up in both training and test

**Wrong window boundary**

the window includes events after the reference point

**Parameters over the whole set**

the mean is computed before the split

**Duplicates and copies**

one record lands in different subsets

**Unavailable feature**

incident triage outcome used as a feature

**Tuning on test**

settings are chosen on the test set

Result of leakage: a good score in evaluation and poor performance in production

# Random vs time-based split

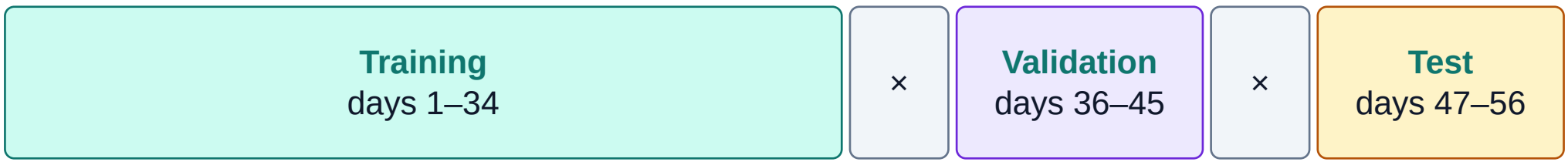
| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Attacks | | | | | | A | A | A | | | | | | | | B | B | B | | |
| Random | Tr | Tr | Te | Tr | Tr | Tr | Te | Tr | Tr | Tr | Te | Tr | Tr | Tr | Tr | Tr | Te | Tr | Tr | Te |
| By time | Tr | Tr | Tr | Tr | Tr | Tr | Tr | Tr | Tr | Tr | Tr | Tr | Tr | × | Te | Te | Te | Te | Te | Te |

Tr — training  Te — test  × — gap

Random: both attacks (A and B) are in training and in test — the model “remembers” them, the score is inflated

By time: A only in training, B only in test — the model faces a new attack, as in real operation

# Subsets and the gap




time → · not to scale

**Training**

fitting model parameters

**Validation**

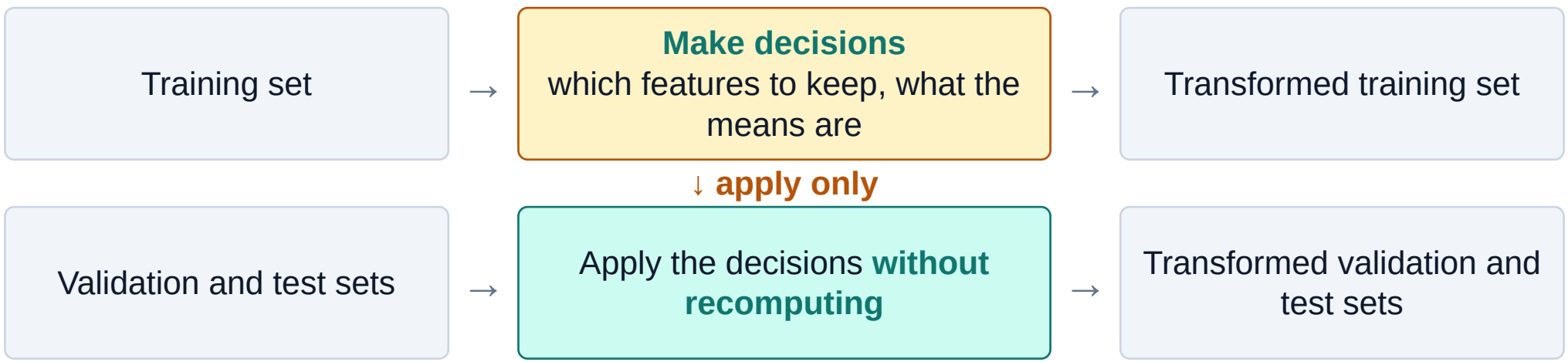
tuning settings and threshold

**Test**

final evaluation — once

> The gap is no shorter than the longest feature window

# Preparation without leakage




Decisions are made on the training set only

# What is computed where

### Whole dataset

schema check
deduplication
normalization
masking
features from the past

### Training only

feature selection
mean values
class rebalancing

### Validation and test

validation: settings and threshold
test: final evaluation, once

# 5

# Labels and label noise

where the answers come from

# Where labels come from

| | |
|---|---|
| **Rule outputs** | ⚠ the model learns to repeat the rules |
| **Analyst triage** | ⚠ only what reached triage is labeled |
| **Incident reports** | ⚠ imprecise start and end times |
| **Reproduced attacks** | ⚠ may differ from real ones |
| **Public datasets** | ⚠ contain errors, a different system |

# Label shift in time

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Actual attack | | | | ● | ● | ● | | | | | | |
| Label in the log | | | | | | | ● | ● | ● | | | |
| Result | | | | ✗ | ✗ | ✗ | ✗ | ✗ | ✗ | | | |

Objects 4–6: a real attack is labeled normal

Objects 7–9: labeled “attack”, but the attack is already over

# Dealing with label noise

### Spot checks

an analyst re-checks ambiguous objects and a random sample

### Gap around an incident

objects near the incident boundary are excluded from training

### Soft labels

a probability of attack instead of “0” and “1”

### Confident learning

labels that contradict model predictions are re-checked

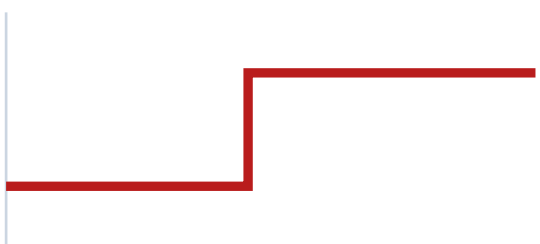
6

# Data drift

data changes

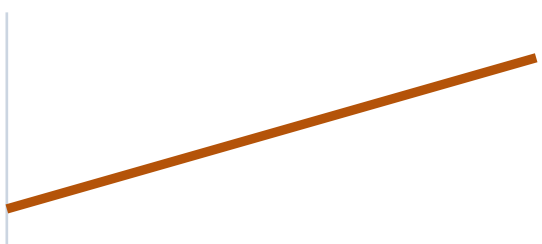
# Kinds of drift

## Sudden



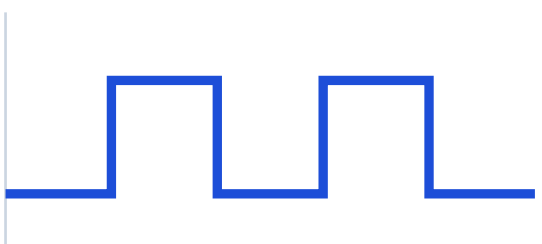
application update

## Gradual



growing number of users

## Recurring



weekdays and weekends

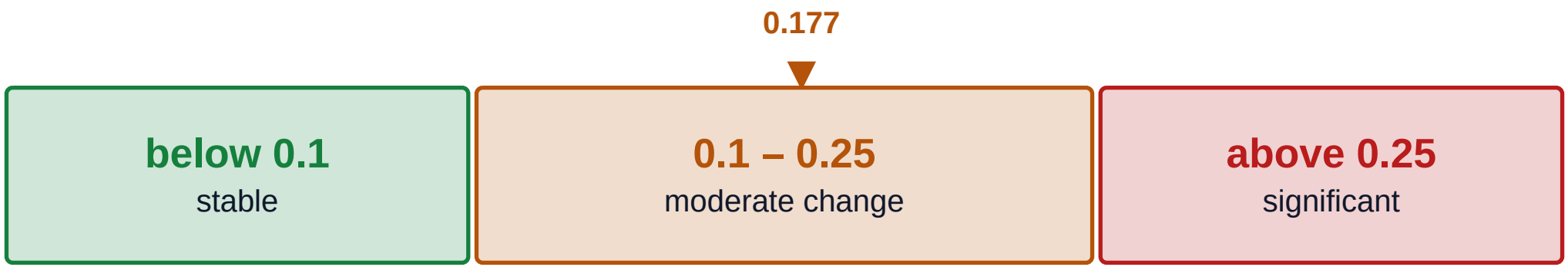
# PSI: comparing distributions

$$\mathrm{PSI} = \sum_i (p_i - q_i) \ln\left(\frac{p_i}{q_i}\right)$$

q — share of objects in the training data, p — in the current data

| Attempts | q | p | p − q | Contribution |
|---|---|---|---|---|
| 1–2 | 0.70 | 0.50 | −0.20 | 0.067 |
| 3–10 | 0.20 | 0.30 | +0.10 | 0.041 |
| more than 10 | 0.10 | 0.20 | +0.10 | 0.069 |
| **Total** | 1.00 | 1.00 | — | **0.177** |

# How to read PSI




Thresholds are conventions, not statistical tests: the choice of bins affects the result

Also: the Page–Hinkley test for data streams; a drop in Precision and Recall on fresh labeled data is the most direct but lagging signal

# Responding to drift

### Scheduled retraining

regularly on fresh data

price: cost of training and validation

### Incremental learning

updating as data arrives

price: error accumulation, “poisoning”

### Revising thresholds and rules

adjusting to new behavior

price: requires an expert

### Manual review

an analyst checks the model

price: delayed reaction

# Key points

| | |
|---|---|
| **1 Dataset** | Objects, features, labels |
| **2 Data quality** | Check the data before training |
| **3 Features** | Computed from already known events |
| **4 Time-based split** | By time with a gap, no leakage |
| **5 Labels and noise** | Check the labels, do not trust them blindly |
| **6 Drift** | Data changes — measure the drift |

Lecture 4

# Preparing Data for Analytical Models

Data quality · features · time-based split · drift